

Math 1 • C21 / D22 re-replace review

Only C21 and D22 after ejecting rational-inequality stems. For checking - not a full paper.

C21 • REPLACE AGAIN

prior pick reduced to rational non-negativity; beyond M1 linear inequalities (M4.17)

The function $f(x) = x^3 + kx$, where k is a constant, takes the same value at $x = a$ and $x = b$. Given that a and b are distinct and are the roots of $x^2 - 4x + 2 = 0$, what is the value of k ?

- A -14
- B -12
- C -10
- D -4
- E 14

D22 • REPLACE AGAIN

explicit rational polynomial inequality; out of Math 1 spec

A rectangle has vertices at the origin O, and at the points A(x , 0), B(x , y) and C(0, y), where x and y are positive integers.

The area of the rectangle OABC is 360.

The gradient, m , of the diagonal OB satisfies the equation:

$$4x - \frac{45}{m} = 30$$

Given that $m < 1$, what is the perimeter of the rectangle?

- A 38
- B 42
- C 76
- D 78
- E 84
- F 92